Software Process Management

Lecture 8

Process Representations in Organisations

- Two major alternatives of making process information publicly known in an organization:
  - Process handbooks and
  - EPGs – Electronic Process Guides
- Both can serve various purposes and have their specific strengths.

Process Handbooks

- Process handbooks are generally thought of as information that helps process performers do the right thing at the right time.
- As such, they define:
- the work that should be done,
- expected outcomes,
- the responsibilities of the various roles,
- the conditions that control the sequencing of the work,
- the conditions that can be used to track progress, etc.

Purpose and Content

- Process handbooks are usually organized according to a logical decomposition of the work that must be done, with chapters for major sub-processes, sections for major activities within each sub-process, and subsections for specific tasks.

- The content and organization of a process handbook should reflect task-related questions posed by process performers, i.e., agents who are assigned to a project and responsible for carrying out the process.

- Some typical questions are:
- – What do I have to do to perform an activity?
- – What are the prerequisites for performing a task?
- – What do I have to produce during the course of my work?
- – Why do I have to perform a task in a specific way (rationale)?
- – Which tasks am I responsible for?
- – How can I determine whether or not I have successfully completed a task?
- – Where can I find a particular template?

- Across all the roles in a process, there will be many questions of many different kinds.
- For each kind of question, there should be one or more views that help in answering questions of this kind.
- For example, an activity decomposition view helps to answer questions about the interdependencies among activities and an artefact lifecycle view helps to answer questions about when to produce which document, and whom to send it to.

- One handbook often cannot conveniently meet the needs of all the roles that will have questions about the process and its performance.
- Actually, many handbooks are needed, each oriented towards some role's domain of questions.
- This severely complicates the maintenance of an organization's process descriptions because changes and additions have to be accurately and consistently made to a possibly large number of handbooks.
- Therefore, a mature organization typically maintains only one comprehensive process model, and generates process handbooks for specific audiences out of this, covering the views required by the respective audience.

The content of Process Handbooks

- – A description of the activities a process performer has to execute,
- – A description of the work products that are produced by each activity, and thus have to be created by the process performer, and
- – A description of the roles associated with the activities.



- In addition to this, there is a variety of optional information that can be covered by process handbooks, depending on their purpose.

- – A description of resources or tools to be used during the activities,
- – Document templates for the work products,
- – Examples of the work products, i.e., an exemplary, “perfect” instance of a particular work product,
- – “How-to”s for specific tasks, e.g., detailed instructions on how to complete the task using a specific tool, and
- – Tips and tricks, e.g., workarounds for a bug in a tool.

- Process handbooks always address a specific audience and try to convey specific information.
- Keep this in mind when developing a process handbook!

Outline and Structure

- The structure of process handbooks varies greatly. So far, there exists no common standard for process handbooks, i.e., no specification on where to find which information.

- Typically, an organization's process handbooks also follow the respective organization's corporate design, which may make the same content look very different across different organizations.

Let's write a process handbook

- Main sections are:

- Introduction
- Definitions
- Process Views
- List of Figures/Tables and Index
- List of References
- Appendices



Introduction

- **Purpose**: What is the purpose of the process handbook—why was it created, e.g., for auditing purposes or for daily usage?
- **Target audience**: Which audience is addressed by the handbook, e.g., programmers or project managers?
- **History**: What is the history of the handbook, which previous version does the current handbook replace, and what was changed?
- **Process specifics**: What is specific about the process that is described in the handbook? Where does it differ from standard/other processes? What makes it special, what should people direct special attention to?

- **Applicability**: In which context(s) is the process applicable, e.g., is it for the development of safety-critical embedded systems or for the development of a Web application?
- **Relationships to other models**: Which other standards (e.g., international standards or other organization standards) or process models is this process handbook derived from? Where does it have similarities, where is it different?
- **Version**: Which version of the process model is described in the handbook? What are the differences to its predecessor/which problems were addressed with this version?
- **Structure**: What is the structure of the handbook? Which information can be found in which section?

Definitions

- **Terminology**: A definition of the terms used in the handbook. This is especially important since different terms are often used synonymously throughout the process community, while other terms have different meanings in different contexts.
- **Abbreviations**: Every organization has its own organizational slang, especially when it comes to abbreviating terms. Therefore, it is important to define the abbreviations used in the handbook.
- **Schemes for describing products, processes, roles, and their relationships:** These descriptions typically follow some kind of (semi-) formal structure. For example, a role can be linked to an activity (“performs”) or to a document (“responsible for”). These schemes should be explained in order to enable the reader to understand the relationships described in the handbook.

Process Views

- The views displayed in the handbook depend on its purpose and content; therefore, no generally applicable recommendation exists. However, typical views contain:
- – **Major entities**: Which are the (sub-) processes/activities/tasks, which are the work products of the processes? Which roles exist, and which resources are used?
- – **Control flow**: Which activity is preceded by which other activity, and which activity follows?
- – **Product flow**: Which activity produces which work product, and which ones does it consume or modify?
- – **Role-specific views**: Which role is participating in which activities? Which role is responsible for which work products?
- – **Hierarchical decomposition**: Which activities and tasks belong to which sub-process?

Other components

- Following have the usual meaning
- List figures/tables and index
- List of references
- Appendices

- Importantly Appendices may contain document templates, examples, checklists, tips and tricks, information on how to tailor the process for specific contexts, etc

Usability of PH

- One of the most important—and difficult to achieve—aspects when it comes to process handbooks is their usability.
- Process handbooks are created for a purpose, a part of which is usually to support process performers in their daily work.
- If they do not like the process handbook, for whatever reason, they are less likely to fully apply the described processes, leading to compliance problems and possibly further mayhem.
- Therefore, the process handbook should make it easy to use it and encourage people to take a look at it.

- While every process handbook is different, there are some general directions that should be considered when creating such a handbook:
- – **Consistency**: The handbook should be consistent. This means that the same terms should always mean the same thing, and this thing should not be described using different terms. It should also be consistent on a higher level of abstraction.
- This means, for example, that when a document is created at some point and a detailed description of the document is referenced, this reference should exist and describe the right document.
- Of course, the underlying process itself should be consistent, too, i.e., when a document is created, it should also be used somewhere else.

- **Up-to-date-ness**: The handbook should be up to date, i.e., it should describe the currently active process. For example, documents that are not used anymore should not be demanded by the process handbook.
- **Design**: The display of information should be functional, yet look nice. For example, the new 8-point company font may look cool, but it may be very hard to read—and thus should not be used for major content.
- The same applies to colours: The extensive use of colours may look great on screen—but if most people print the handbook on black-and-white printers and the colours suddenly all look like the same grey, this does not help.

- Accessibility: The information in the process handbook should be readily accessible. A single printed version of the handbook on the department's bookshelf is not likely to be used—or applied during daily work. An EPG, prominently linked in the organization's intranet or process portal, with powerful search functionality and direct template download, is much more helpful.

- Remember: The best process handbook is worthless if it is difficult to use and is thus ignored!

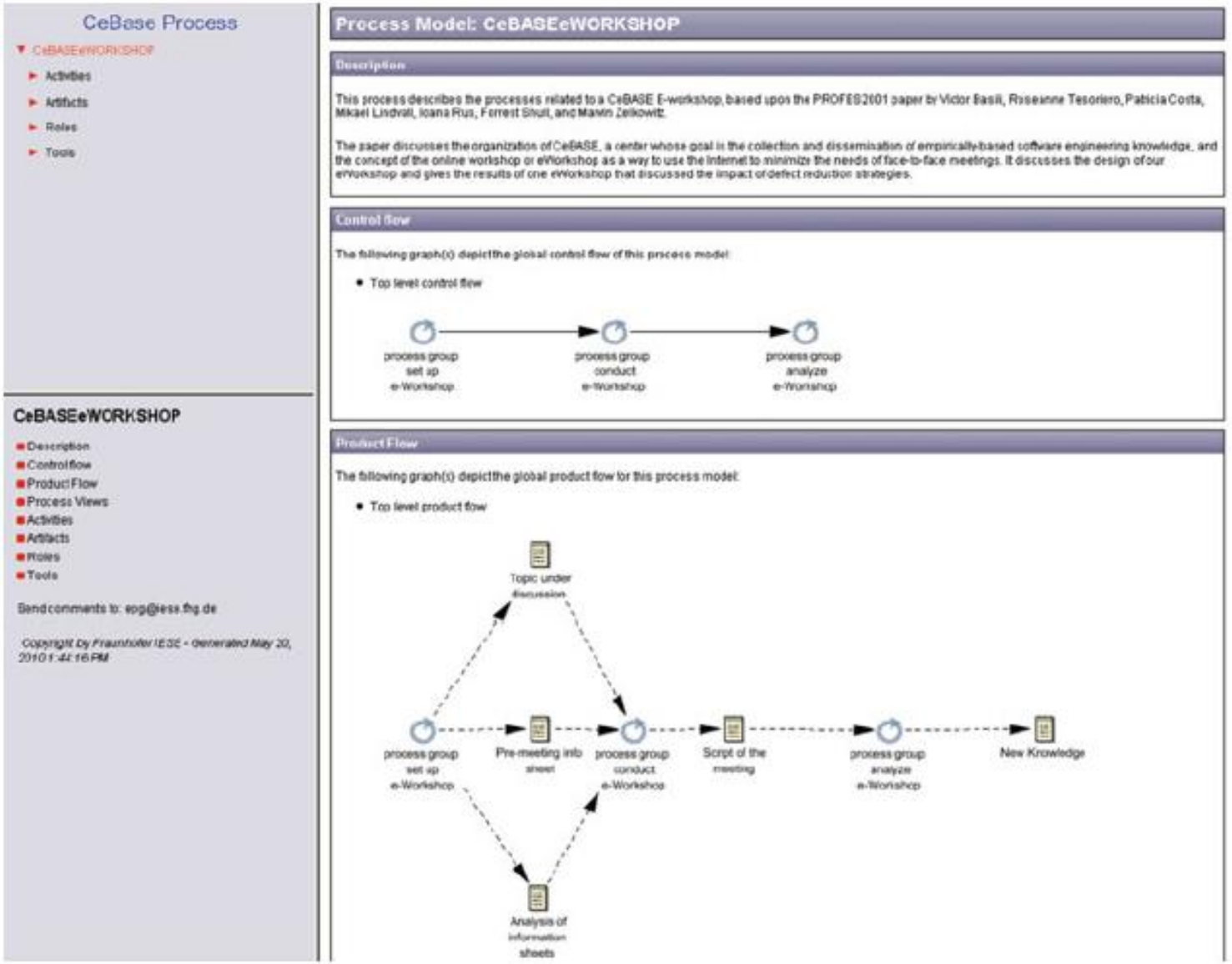
Electronic Process Guides

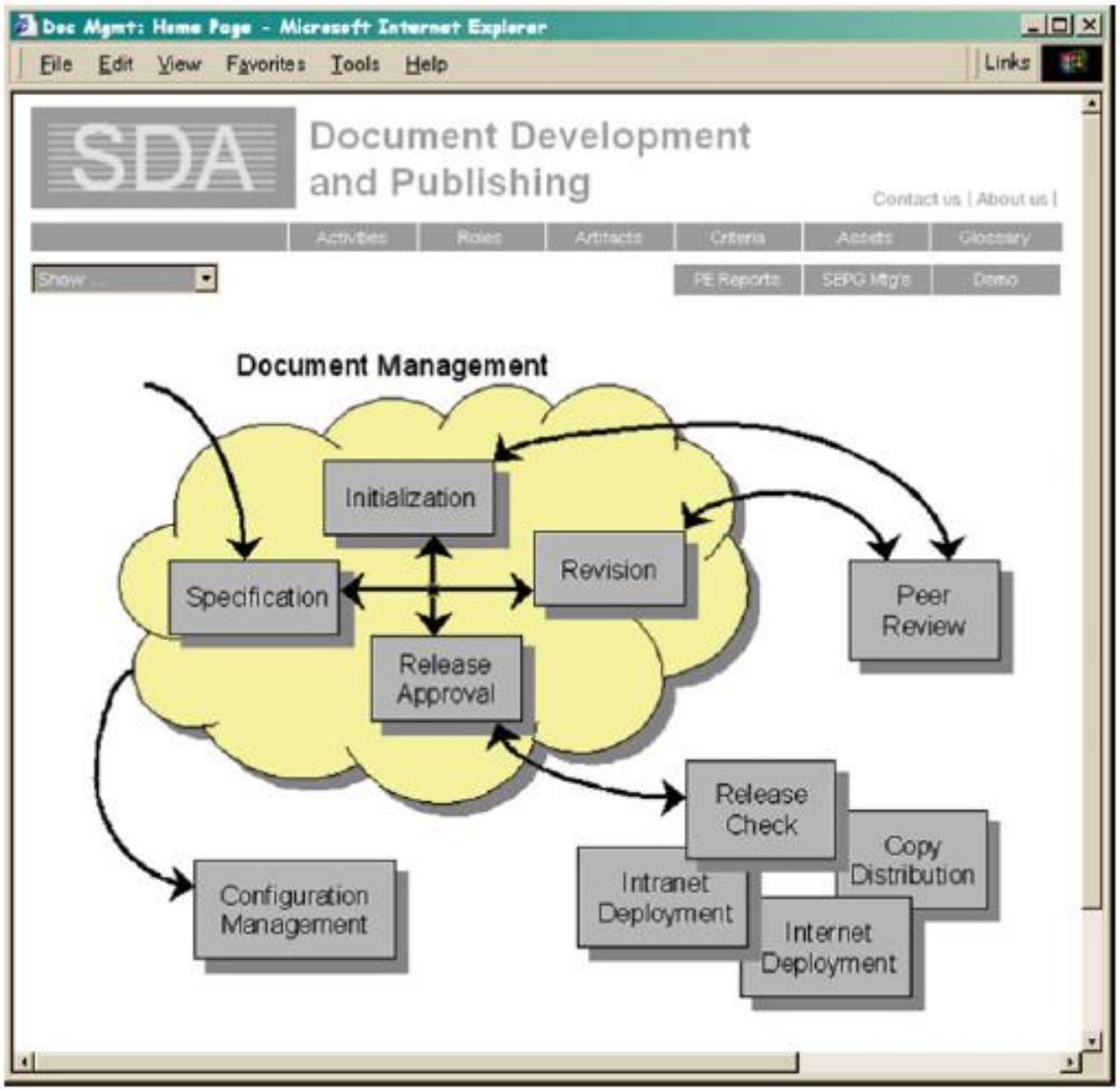
- EPGs are typical instruments used to guide process performers in doing the right thing at the right time—much more so than process handbooks.
- While a process handbook as a linear representation of the process cannot really capture its networked structure, an EPG can.
- Interlinked Web pages allow following every possible path through the process easily, and are thus much easier to use for process guidance than a handbook that either requires following the given structure, or necessitates a lot of skimming through the pages.

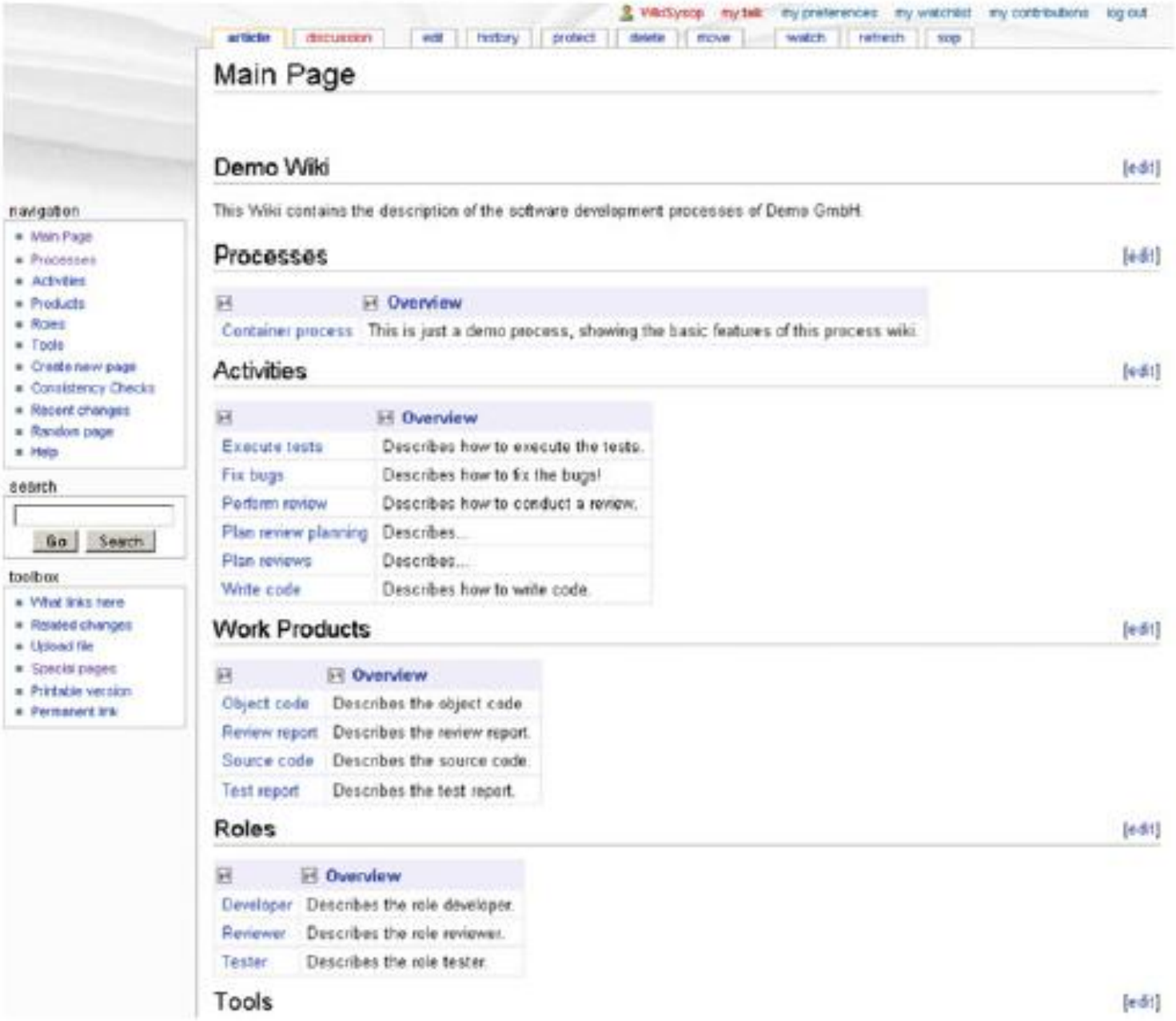
- Given the purpose of guidance for process performers, the content may be similar to that of process handbooks, yet a lot more useful for performing the process.
- For example, a template for a specific work product may be provided for download from the intranet—so that the creation of, for example, a requirements specification as prescribed is a lot simpler.
- A process handbook will typically be less comfortable to use and require, for instance, going to a specific intranet Web site manually (by typing in the URL).
- While this may sound like a minor issue, it may just be the reason for people not to use the provided template.

- An EPG may also support process performance by providing different views on the process.
- While a process handbook typically comprises the complete process (and thus features a rather global view), an EPG may provide different views for different roles.
- For example, a tester might not always need to know how to use the requirements management tool—thus, for his view, such details may be omitted.

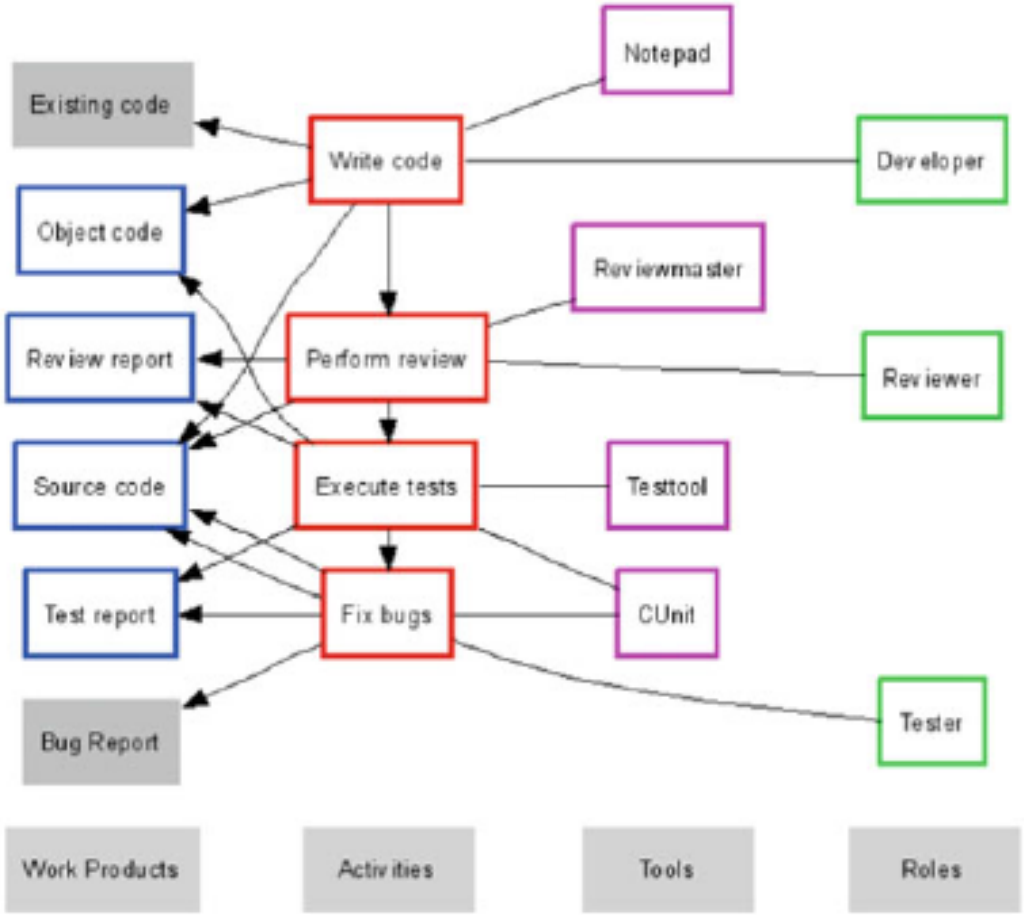
EPG Example


CeBase Process
CeBASEeWORKSHOP
Activities
Artifacts
Roles
Tools
CeBASEeWORKSHOP
Description
Control flow
Product Flow
Process Views
Activities
Artifacts
Roles
Tools
Send comments to: epg@iese.fhg.de
Copyright by Fraunhofer IESE - Generated May 20, 2010 1:41:16 PM
Process Model: CeBASEeWORKSHOP
Description
This process describes the processes related to a CeBASE E-workshop, based upon the PROFES2001 paper by Victor Basili, Roseanne Tesoriero, Patricia Costa, Mikael Lindvall, Ioana Rus, Forrest Shull, and Marvin Zelkowitz.
The paper discusses the organization of CeBASE, a center whose goal is the collection and dissemination of empirically-based software engineering knowledge, and the concept of the online workshop or eWorkshop as a way to use the Internet to minimize the needs of face-to-face meetings. It discusses the design of our eWorkshop and gives the results of one eWorkshop that discussed the impact of defect reduction strategies.
Control flow
The following graph(s) depict the global control flow of this process model:
Top level control flow
process group set up e-Workshop
process group conduct e-Workshop
process group analyze e-Workshop
Product Flow
The following graph(s) depict the global product flow for this process model:
Top level product flow
Topic under discussion
process group set up e-Workshop
Pre-meeting info sheet
process group conduct e-Workshop
Script of the meeting
process group analyze e-Workshop
New Knowledge
Analysis of information sheets


Doc Mgmt: Home Page - Microsoft Internet Explorer
File Edit View Favorites Tools Help
Links
SDA
Document Development and Publishing
Contact us | About us |
Activities
Roles
Artifacts
Criteria
Assets
Glossary
Show ...
PE Reports
SEPG Mtg's
Demo
Document Management
Initialization
Specification
Revision
Peer Review
Release Approval
Release Check
Copy Distribution
Intranet Deployment
Internet Deployment
Configuration Management


WikiSysop
my talk
my preferences
my watchlist
my contributions
log out
article
discussion
edit
history
protect
delete
move
watch
refresh
sop
Main Page
navigation
Main Page
Processes
Activities
Products
Roles
Tools
Create new page
Consistency Checks
Recent changes
Random page
Help
search
Go
Search
toolbox
What links here
Related changes
Upload file
Special pages
Printable version
Permanent link
Demo Wiki
[edit]
This Wiki contains the description of the software development processes of Demo GmbH.
Processes
[edit]
Overview
Container process
This is just a demo process, showing the basic features of this process wiki.
Activities
[edit]
Overview
Execute tests
Describes how to execute the tests.
Fix bugs
Describes how to fix the bugs!
Perform review
Describes how to conduct a review.
Plan review planning
Describes...
Plan reviews
Describes...
Write code
Describes how to write code.
Work Products
[edit]
Overview
Object code
Describes the object code.
Review report
Describes the review report.
Source code
Describes the source code.
Test report
Describes the test report.
Roles
[edit]
Overview
Developer
Describes the role developer.
Reviewer
Describes the role reviewer.
Tester
Describes the role tester.
Tools
[edit]

Activity Graph [edit]



Inputs [edit]

Here, all process-external inputs will be listed
Someting from outside

Outputs [edit]

Here, all outputs leaving this process will be listed
Something going outside

Deploying Prescriptive Process Models

- Process deployment deals with getting a prescriptive process into daily practice, i.e., getting people to follow the new or changed process.
-
- This means changing people's behaviour, which is a difficult task in any context.
- Two general strategies for deploying a changed process to an organization are the big-bang and the phased approach.

Big Bang

- The complete organization is switched from the old to the new process at the same time.
- In this case, there will be no confusion resulting from people working in different projects/departments using different processes.
- However, this strategy requires a large amount of support capability for educating and coaching the employees. An intensified version of the big-bang strategy also switches running projects to the new process.
- However, this creates a lot of overhead; therefore, this is rarely done.

Phased

- The new process is applied only selectively in specific, newly starting projects.
- This requires only limited training and coaching capabilities and eliminates the switching overhead.
- However, if this strategy is pursued on a continuing basis, there will be a plethora of process variants active within the organization at the same time.

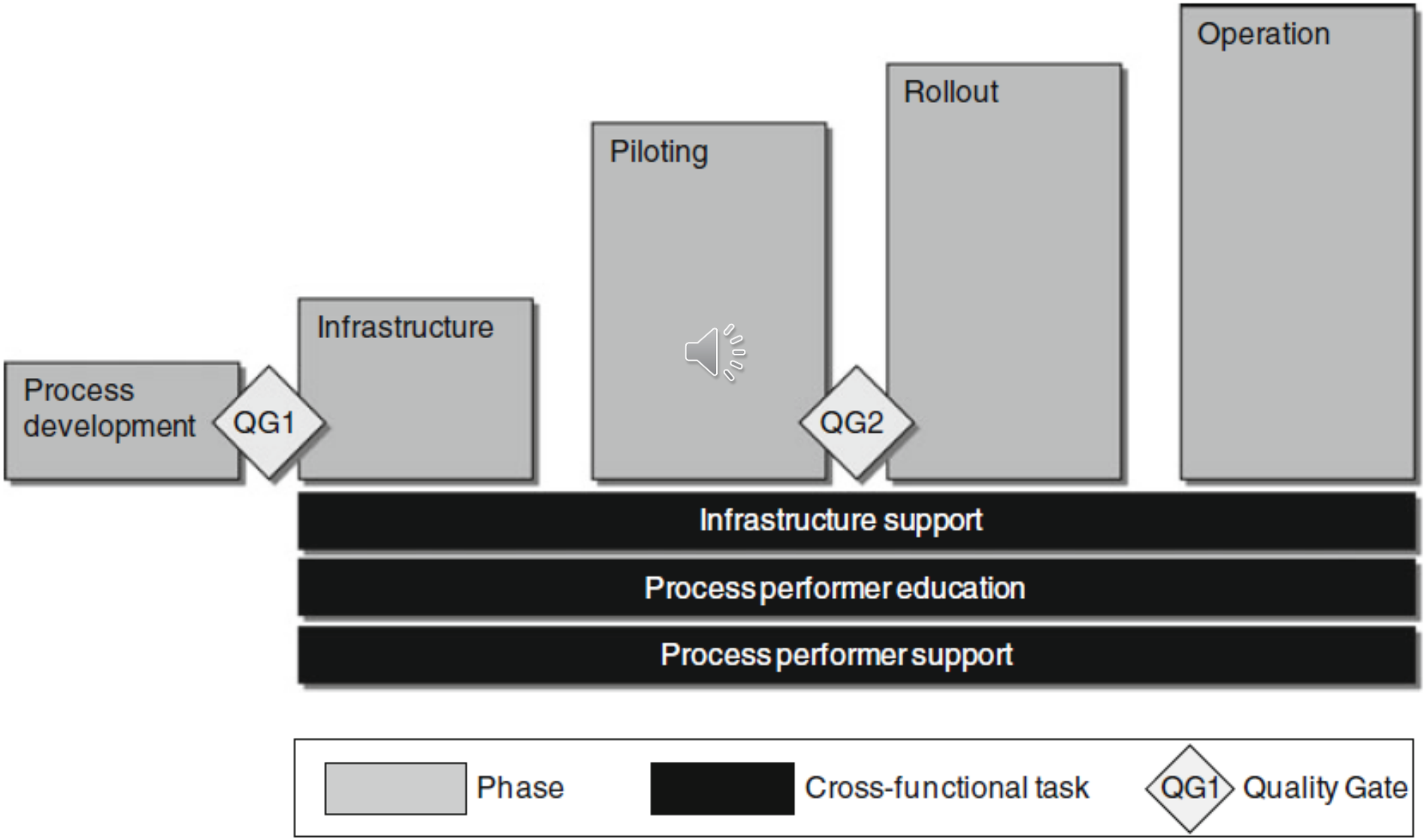
An Exemplary Deployment Approach

- With every change to a process, there is a risk that the changed process does not yield its expected benefits, i.e., does not perform better than the old one, or even worse, does not even perform as well as the old process and thus makes the situation worse.



- Therefore, a changed process should be tested before it is widely applied.

- A staged strategy permits to do this and has, in fact, proven to be a good choice.


Operation
Rollout
Piloting
Infrastructure
Process development
QG1
QG2
Infrastructure support
Process performer education
Process performer support
Phase
Cross-functional task
QG1
Quality Gate

Process development

- During this phase, the changed process is developed, i.e., its model is created and the necessary documentation (handbooks, EPGs, etc.) is prepared.
- A typical approach is to create a descriptive model of the current process, analyse it for weaknesses, and modify the model to remove these weaknesses.
- However, this activity is not considered part of process deployment, and therefore is not detailed here.

Quality Gate 1: Piloting Maturity

- This quality gate checks the changed process to see whether it is mature enough to be piloted. In particular, the following aspects should be checked:

- – Goal achievement 
- Are the goals of the changed process documented in a verifiable form?
- Which problems are addressed?
- Does a cost/benefit analysis justify changing the process?

- – Process handbook (presentation and contents)
- Are the essential elements included?
- Are the presentation and contents okay for the target audience?
- Is the process handbook complete and correct?

- – Quality assurance
- Was the changed process reviewed? By whom? Was the review documented?
- Was the process reworked according to the review results?
- Were all stakeholders identified and involved?

- – Context factors
- What is the environment (technical/social) of the target audience?
- Are document templates available?
- Is there a definition of when the process is “successful”? What are the success criteria?
- Has a deployment strategy been defined?

Infrastructure

- The infrastructure necessary to adequately support the new process must be supplied before the piloting starts, in order to provide realistic circumstances.
- This includes providing a sufficient number of tools (e.g., a requirements management tool), workplaces, hardware resources such as engine test beds or test vehicles, as well as education resources such as classrooms, personnel for helpdesk services, and others.

Process Performer Education and Support

- These tasks are cross-functional, i.e., they are required to run continuously during the deployment of the changed process.
- At the beginning, education will require more resources, while at the end, this will shift towards supporting the process performers in their daily work.
- In particular, the following aspects should be considered:
- - Process performer empowerment
- Create trust and acceptance
- Enable participation of process performers: include their input, let them take responsibility

– Education before project start

- Common introductory courses
- Role-specific courses
- Tool-specific courses

- Process performer support before project start, during project runtime, and after the project finishes

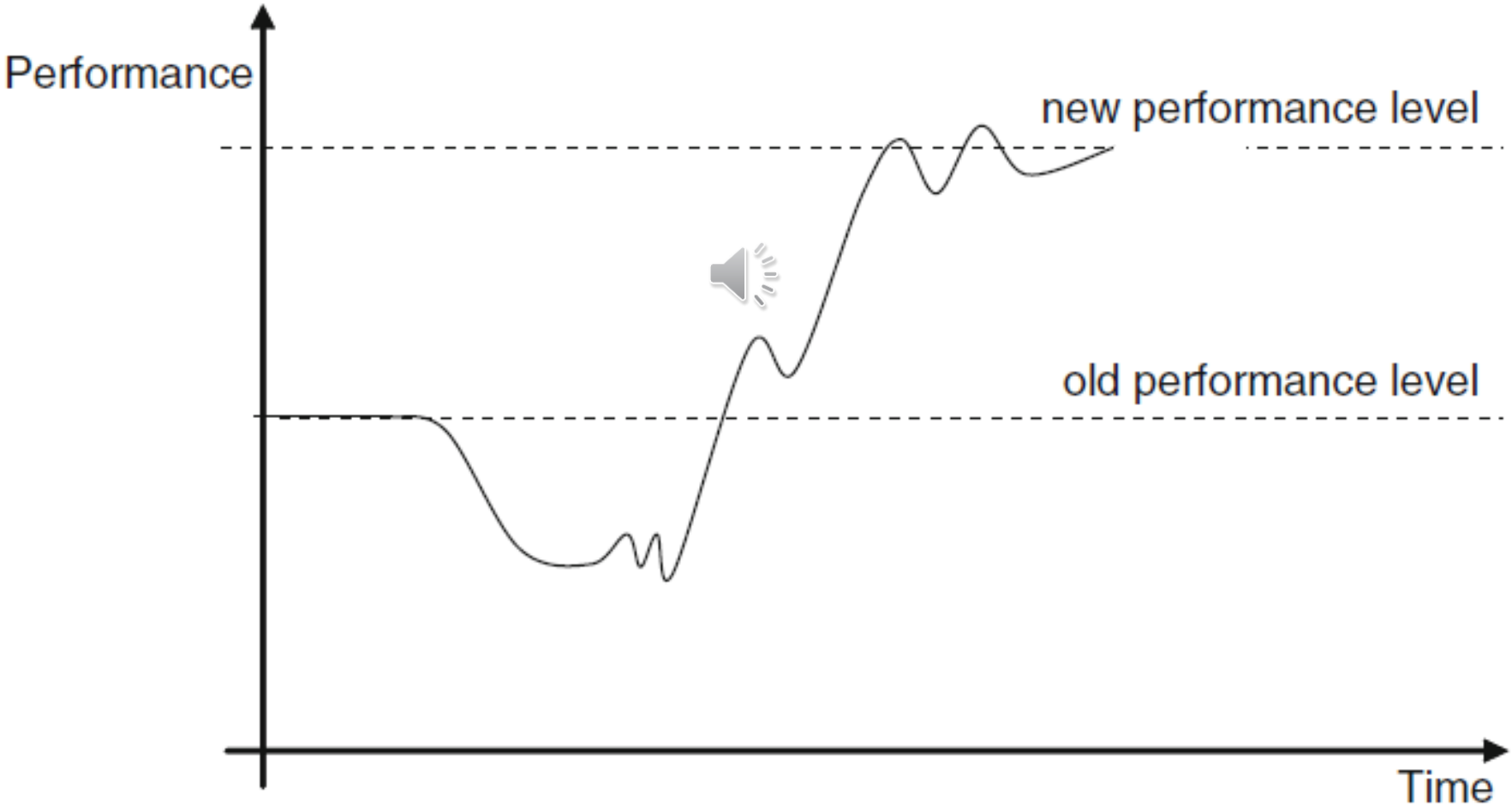
- Provision of education materials
- Electronic newsletters
- Community building: forum, wiki, blog
- Coaching of process performers during the project
- Provision of helpdesk service

- – Feedback mechanisms before project start, during project runtime, and after the project finishes
- To decrease the process performers' fear
- To identify process problems ("you missed a step here")
- To identify potentially problematic areas ("this is never going to work")
- To check when the project is over: Was the education adequate? Tool support? Process handbook?
- To create a "Wishlist" for future changes to the process

Piloting

- When Quality Gate 1 has been successfully passed, the pilot project(s) using the changed process can commence.
- During a pilot project, the changed process is tested, i.e., it is applied in a realistic setting and the effects of its application are recorded.
- The goal of such a pilot project is to evaluate whether the changed process is fit for organization-wide deployment.
- However, it should not be forgotten that the pilot project itself introduces a bias: By choosing “simple” or “complicated” projects, the results can be manipulated.

- Therefore, a representative project should be chosen.
- In addition, it is also likely that project performance (e.g., cycle time) will decrease at the beginning.
- This is because a changed process always introduces a certain amount of confusion, which degrades process performance.
- It is important not to mistake the confusion-generated performance decrease with a general process failure; however, making this distinction is not easy and requires a certain experience.
- Figure shows the typical performance development over time when deploying a changed process, with a temporary decrease compared to the old performance level, and subsequent increase and stabilization on the new, higher performance level.


Performance
new performance level
old performance level
Time

Quality Gate 2: Rollout Maturity

- If piloting of the changed process was successful, it should be evaluated whether the new process has actually been applied—otherwise an untested process could be rolled out organization wide.
- Such an evaluation typically analyses those artefacts that should be prepared according to the changed process and/or interviews the process performers who applied the changed process.
- It should also be evaluated whether it is plausible that the changed process can also reach its goals outside the pilot project.

Rollout

- The rollout takes the changed process to the entire organization.
- Whether to choose a more big-bang-like rollout or a phased rollout depends on various influence factors within the organization, including, but not limited to, factors like the degree to which processes and projects are interwoven, the available support resources, the change climate, and many others.
- Typically, it is not a good idea to switch running projects—however, this may be necessary for heavily interrelated projects. This must be decided on a case-by-case basis.

- *First change the process organization, then the organizational structure.*
- This is suitable for small process changes, or, in a phased rollout scenario, in the early phases (when only few process performers follow the new process).
- If the future of the changed process is unknown (i.e., if it is unclear whether it will remain active), it is also a good idea not to change the organizational structure too early.

- *First change the organizational structure, then the process organization.*
- This is advisable if the new process mandatorily requires certain services or structures within the organization, e.g., a test centre or specific hardware.
- In a big-bang rollout scenario, this may also be the way to go.
- *Concurrent introduction of new organizational structure and process organization.*
- This is, in fact the most frequently chosen variant. It reduces additional friction losses due to improper organizational structures and allows for the creation of the new organizational structure as needed, thus minimizing overall effort.
- However, it may increase the effort required for rolling out the process compared to the organizational-structure-first variant.

Operation

- When the changed process has been rolled out to the entire organization, the operational phase starts.
- During this phase, the process should be closely monitored and evaluated with respect to its effects.
- However, this activity is not considered part of process deployment, and therefore not discussed here.

Experience from Industrial Practice

- In today's software organizations, process deployment is often performed less than optimally.
- Typical process management failures thus include the following:
- – Process handbooks do not fulfil their purpose, e.g., important information is missing or they are too detailed to read.
- – Process definitions are vague and incomplete, i.e., the actual problems are not addressed, but hidden underneath soft, meaningless statements.
- – Interteam interfaces are poorly defined.

- – Non-application of the changed process is not sanctioned, i.e., people are not forced to use the changed process and thus do not use it.
- – The process engineers are not seen as service providers who assist the developers in their daily work, but rather as enemies of success, i.e., only keeping developers from “doing the real work.”

- There are more problems, of course.
- However, these are the most common ones that should be considered in any process organization.